SHOC SOPs 3

Emergency Telecommunications SOPs

Communication infrastructure, network continuity, and emergency communication protocols.

SADC Humanitarian Emergency Operations Centre, Nacala, Mozambique

SADC HUMANITARIAN AND EMERGENCY OPERATIONS CENTRE

STANDARD OPERATING PROCEDURE

ICT & Emergency Telecommunications Protocol And Support of Communications Network

Aligned to SADC ICT Protocol 2026 | UNDAC Handbook 8th Edition (2024) | ERT SOP

IFRC ERU SOPs | ETC Field Standards | FEMA NIFOG | Sendai Framework

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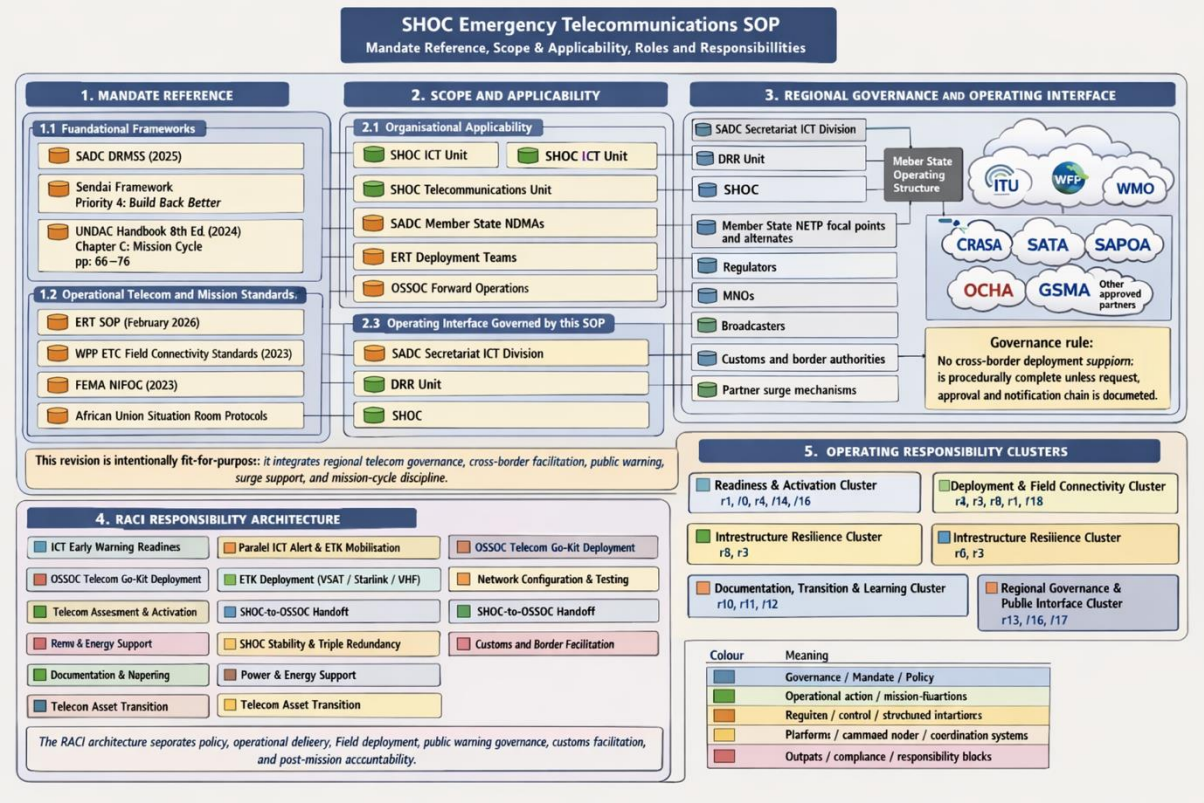
1. Purpose and Objective

To establish, operationalise, and maintain world-class emergency telecommunications infrastructure across all phases of the UNDAC mission lifecycle, from continuous pre-crisis readiness through mission-end transition and lessons learned.

The SOP ensures resilient, interoperable, and redundant communication systems linking the SHOC Situation Room, NDMAs, Member States, the AUC Situation Room, forward-deployed OSSCOs, and ERT field teams. It embeds telecommunications within the early warning architecture as a proactive readiness function rather than a post-activation logistics process, and prescribes parallel ICT activation synchronised with ERT mobilisation workflows.



2. Mandate Reference

This fit-for-purpose revision further aligns the SOP to the SADC Model National Emergency Telecommunications Plan (NETP), SADC Framework for Harmonisation of Radio Frequencies for the Public Protection and Disaster Relief (PPDR), EW4All implementation requirements, Tampere Convention, SADC/CRASA Emergency Services Numbering harmonisation work, and the regional governance and request-routing expectations validated through the SADC Secretariat emergency telecommunications process.

3. Scope and Applicability

This SOP applies to SHOC ICT and Telecommunications Units, SADC Member State NDMAs, ERT deployment teams, OSSOC forward operations, and partners during activation of the SADC Regional Response Mechanism. It covers all six UNDAC mission lifecycle phases: Awareness (continuous readiness), Alert (parallel activation), Standby (team and kit

preparation), Dispatch (deployment), On-Mission (operations), and Mission-End (transition, demobilisation, and lessons learned). It additionally governs the operating interface among the SADC Secretariat Infrastructure Directorate (ICT Division), DRR Unit, ICT Unit, SHOC, Member State NETP focal points and alternates, National Emergency Telecommunications Working Groups (NETWGs), regulators, MNOs, broadcasters, customs and border authorities, and partner surge mechanisms where emergency telecommunications support is requested or augmented.

4. Roles and Responsibilities

For activation, support, and transition activities, SHOC shall operate within a defined regional governance architecture. The SADC Secretariat ICT Division provides regional policy and institutional coordination; the DRR Unit validates disaster-support coordination pathways; the ICT Unit provides ICT/Network Equipment; SHOC provides operational telecom planning, deployment support, and mission assurance; Member States act through their NDMA, NETP focal point, regulator, and NETWG structures; CRASA, SATA, and SAPOA provide sector coordination support where relevant; and ITU, WFP, WMO, IFRC, OCHA, GSMA, and other approved partners may provide technical, training, mapping, or surge support. No cross-border deployment support shall be treated as procedurally complete unless the responsible request, approval, and notification chain has been documented.

Function	Responsible	Accountable	Consulted	Informed
ICT Early Warning Readiness	SHOC ICT Specialist	ICT Manager	Situation Room, ERT Coordinator	AUC Situation Room
Parallel ICT Alert & ETK Mobilisation	ICT Specialist / ERT Coordinator	ICT Manager	NDMAs, ITU	SHOC Ops Centre
OSSOC Telecom Go-Kit Deployment	ICT Specialist (ERT-deployed)	ERT Field Lead	NDMA Host, SATCOM Providers	SHOC NOC
Telecom Assessment & Activation	ICT Office	SHOC Executive Director	ITU, WFP, MNOs	AUC Situation Room
ETK Deployment (VSAT/Starlink/VHF)	SHOC ICT Team	ICT Manager	NDMAs	Member States
Network Configuration & Testing	Telecom Engineers	ICT Manager	SATCOM Providers	SHOC Ops Centre
SHOC-to-OSSOC Handoff	ICT Specialist	ERT Field Lead / ICT Manager	SHOC NOC, NDMA	AUC Situation Room
SHOC Stability & Triple Redundancy	NOC / SOC Teams	ICT Manager	Cloud Admin, Cybersecurity	Executive Director
Power & Energy Support	Logistics Unit	Ops Director	UNDP	FANR
Documentation & Reporting	IM Officer	ICT Manager	MEL Team	SHOC Secretariat
Telecom Asset Transition	ICT Specialist	ERT Field Lead	NDMA Host	SHOC MEL
Telecom AAR & Lessons Learned	ICT Specialist / MEL Officer	ICT Manager	ERT Coordinator	SHOC Secretariat
Regional Governance Coordination	SADC Secretariat Infrastructure Directorate (ICT Division) / DRR Unit/ ICT Unit	Executive Director, SHOC	CRASA, SATA, SAPOA, Member State ICT Ministries	Member States / Partners
Member State NETP Focal Point Control	SHOC ICT Manager	Ops Director	NDMA, Secretariat Infrastructure Directorate (ICT Division), DRR Unit, ICT Unit	Situation Room / ERT Coordinator
NETWG Notification & Consultation	NDMA NETP Focal Point	NDMA / SHOC Executive Director	Regulator, MNOs, broadcasters, customs, operators	SADC Secretariat Infrastructure Directorate (ICT Division)
Public Warning Governance (CAP/CB/LB-SMS)	NDMA Alerting Authority / PR Officer	NDMA Competent Authority	Regulator, MNOs, broadcasters, SHOC ICT	Public / at-risk communities

Customs and Border Facilitation	SCM / Legal / ICT Manager	Ops Director	Border authorities, customs, regulator, ITU	ERT Field Lead / Warehouse
External Surge Equipment Augmentation	ICT Manager	Executive Director, SHOC	ITU Harare, WFP/ETC, approved partners	Situation Room / ERT Coordinator

5. Being Aware — Continuous Telecom Readiness

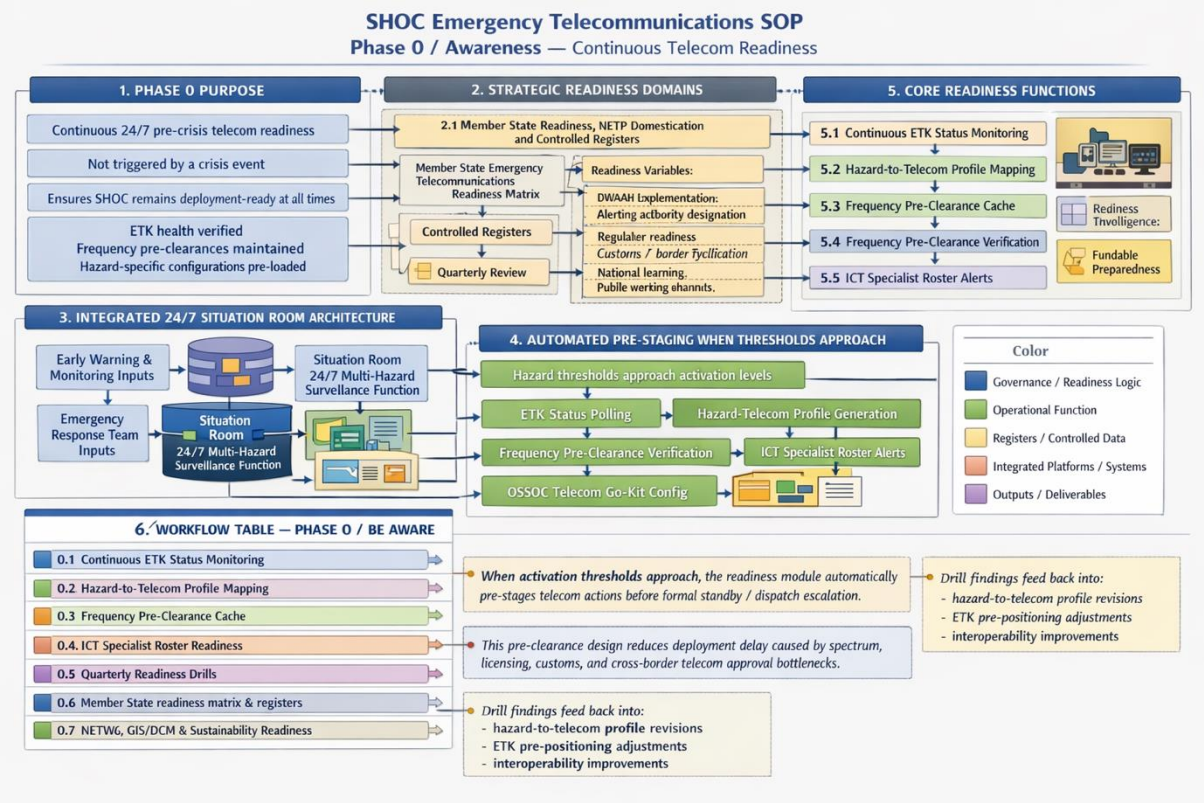


Figure 2: ICT Early Warning Readiness Module — Integration with Situation Room

5.1 Purpose

The awareness and preparedness phase establishes the proactive, pre-crisis telecommunications readiness posture required by the UNDAC Handbook (p. 66), which states that preparedness steps must be completed in advance because the receipt of an alert is not the time to start thinking about what to bring or who to call. This phase operates continuously, 24 hours a day, 7 days a week, and is not triggered by any crisis event. It ensures that SHOC's emergency telecommunications capability maintains deployment-ready status at all times, with ETK health continuously verified, frequency pre-clearances maintained, and hazard-specific telecom configurations pre-loaded.

5.2 ICT Early Warning Readiness Module

5.2.1 Member State Readiness, NETP Domestication and Controlled Registers

As part of continuous readiness, SHOC shall maintain a Member State Emergency Telecommunications Readiness Matrix that records, at minimum, NETP domestication status, EW4All implementation status, alerting authority designation, regulator readiness, customs and border facilitation readiness, national roaming provisions, available public warning

channels, and minimum host counterpart capability. The matrix shall classify each Member State as Mature, Transitional, or Assisted Readiness and shall be reviewed quarterly with the SADC Secretariat Infrastructure (ICT Division) and DRR Unit. Controlled operational registers shall also be maintained for NETP focal points and alternates, EW4All alerting authorities, regulators, MNO emergency contacts, broadcasters, border/customs contact points, and ITU Southern Africa Area Office (Harare) escalation contacts.

5.2.1 NETWG, GIS/DCM, Financing and Sustainability Preparedness

This phase of readiness shall include pre-identification of NETWG activation procedures, GIS layer ownership, and financing sustainability assumptions. SHOC shall maintain a minimum GIS/DCM baseline pack containing vulnerability maps, network coverage maps, radio/TV/mobile/fixed/terrestrial/satellite coverage, EOCs, dams, settlements, corridors, and cross-border logistics nodes. The ICT Manager and Finance focal point shall annually validate capital refresh needs, recurring OPEX lines, airtime and subscription commitments, training budgets, and partner co-financing assumptions so that operational readiness remains fundable and not merely procedural.

The ICT Early Warning Readiness Module is embedded within the Situation Room architecture and operates as an integral component of the 24/7 multi-hazard surveillance function. The module receives automated feeds from myDEWETRA, GDACS/ADAM/PDC, the INFORM Sub-national Risk Index, regional cyclone tracking systems, and NDMA verification channels. When hazard thresholds approach activation levels, the module triggers automated pre-staging actions including ETK status polling, hazard-to-telecom profile generation, frequency pre-clearance verification, parallel ICT specialist roster alerts, and OSSOC Telecom Go-Kit configuration.

5.3 Step-by-Step: Be Aware Phase Procedures

5.3.1 Continuous ETK Status Monitoring

The ICT Specialist, in coordination with the SHOC Warehouse Manager, shall implement and maintain an automated ETK Status Monitoring System that continuously verifies the readiness of all pre-positioned Emergency Telecom Kits. The system shall poll battery charge levels (minimum 80% maintained), firmware currency (auto-update within 72 hours of vendor release), antenna integrity (monthly physical inspection), satellite subscription status (active and funded), SIM card activation status, and GPS tracking beacon functionality. Status data shall feed into the Situation Room ICT Dashboard with automated amber/red alerts for any parameter falling below threshold. The ICT Specialist shall resolve all amber alerts within 24 hours and red alerts within 8 hours.

5.3.2 Hazard-to-Telecom Profile Mapping

The ICT Early Warning Readiness Module shall maintain pre-configured telecommunications deployment profiles for each SADC hazard type. These profiles shall specify optimal ETK configuration (antenna type, bandwidth allocation, redundancy mode), frequency requirements, power system specifications, and OSSOC Go-Kit manifest for the following hazard categories: tropical cyclone, riverine flood, drought, earthquake, epidemic/pandemic, conflict/displacement, industrial accident, and multi-hazard compound event. Profiles shall be reviewed quarterly and updated following each mission based on field lessons learned. Each profile shall be linked to the INFORM Sub-national Risk Index to enable automatic pre-selection when hazard indicators approach threshold levels. Each hazard profile shall also

specify the minimum GIS/DCM layers required, the likely public warning channels to be activated, national roaming contingencies, and whether SHOC-only assets are sufficient or whether early augmentation by ITU Southern Africa Area Office (Harare) or other approved partners should be pre-considered.

5.3.3 Frequency Pre-Clearance Cache Management

The ICT Manager shall maintain a Frequency Pre-Clearance Cache containing pre-arranged generic frequency clearances aligned to SADC Global MOUs, SADC Framework for Harmonisation of Radio Frequencies for the Public Protection and Disaster Relief (PPDR), active radio licenses for SHOC-owned equipment, regulatory contact databases for all 16 SADC Member State spectrum authorities, Tampere Convention documentation packages pre-prepared for each Member State, and template frequency allocation requests requiring only disaster-specific details to be completed. The cache shall be verified monthly and updated within 48 hours of any regulatory change notification. This pre-clearance approach directly addresses Gap 8 by ensuring that frequency allocation processes do not create deployment delays during the Standby and Dispatch phases. The cache shall include customs document packs, temporary admission templates, serialised manifests, re-export templates, regulator escalation ladders, roaming concession precedents, and template letters for formal support requests routed through the SADC Secretariat and ITU Harare where required.

5.3.4 Quarterly Redundancy and Readiness Drills

The ICT Manager shall schedule and conduct quarterly telecommunications readiness drills that simulate the complete activation-to-OSSOC-deployment sequence. Each drill shall test ETK mobilisation speed against the enhanced 8-hour OSSOC connectivity target, triple-redundancy failover performance against the 30-second auto-failover target, parallel ICT alert synchronisation with ERT Portal alerts, OSSOC Telecom Go-Kit deployment including NOC-lite establishment, and SHOC-to-OSSOC handoff protocol execution. Drill results shall be documented in the Quarterly Readiness Drill Report (Annex 3.1-G) and findings shall be incorporated into hazard-to-telecom profile revisions and ETK pre-positioning adjustments. Readiness drills shall also test CAP approval chains, Cell Broadcast or LB-SMS fallback logic where available, NETWG notification, customs facilitation rehearsal, regulator/MNO interoperability calls, amateur radio or community-station fallback, and GIS/DCM product generation.

5.3.5 Be Aware Phase Workflow Table

Step	Activity	How to Perform	Output / Deliverable	Responsible	ETD	SHOC Link
0.1	Continuous ETK Status Monitoring	Implement automated polling of all pre-positioned ETKs: battery, firmware, antenna, subscriptions, SIMs. Dashboard alerts for sub-threshold parameters.	ETK Status Dashboard (live)	ICT Specialist / Warehouse Manager	Continuous (24/7)	Situation Room Dashboard
0.2	Hazard-to-Telecom Profile Mapping	Maintain pre-configured telecom deployment profiles per SADC hazard type. Link to INFORM risk index for auto-pre-selection.	Hazard-Telecom Profile Library	ICT Specialist	Quarterly review	ERT Coordinator
0.3	Frequency Pre-Clearance Cache	Maintain generic clearances, active licences, regulatory contacts, Tampere Convention packages, template requests.	Frequency Pre-Clearance Register	ICT Manager	Monthly verification	Legal Advisor

0.4	ICT Specialist Roster Readiness	Maintain updated ICT specialist availability on ERT Portal. Verify training certifications and deployment readiness.	ICT Roster Status Report	ICT Manager / HR	Monthly	ERT Coordinator
0.5	Quarterly Readiness Drills	Conduct end-to-end telecom activation drill: ETK mobilisation, redundancy failover, OSSOC Go-Kit deployment, handoff.	Quarterly Drill Report (Annex 3.1-G)	ICT Manager / MEL Officer	Quarterly	Executive Director
0.6	Member State readiness matrix & registers	Review NETP status, EW4All, alerting authorities, regulator/customs readiness, focal points and alternates. Update controlled registers and classify readiness level.	Readiness Matrix + Controlled Registers	ICT Manager / Secretariat ICT Liaison	Quarterly	DRR Unit / NDMA Focal Point
0.7	NETWG, GIS/DCM & sustainability readiness	Validate NETWG procedures, baseline GIS/DCM layers, financing assumptions, partner augmentation options, and recurring telecom OPEX commitments.	Preparedness Validation Note	ICT Manager / Finance Focal Point	Quarterly	Executive Director / Secretariat Infrastructure Directorate - ICT Division

6. Phase 1: Alert — Parallel Activation

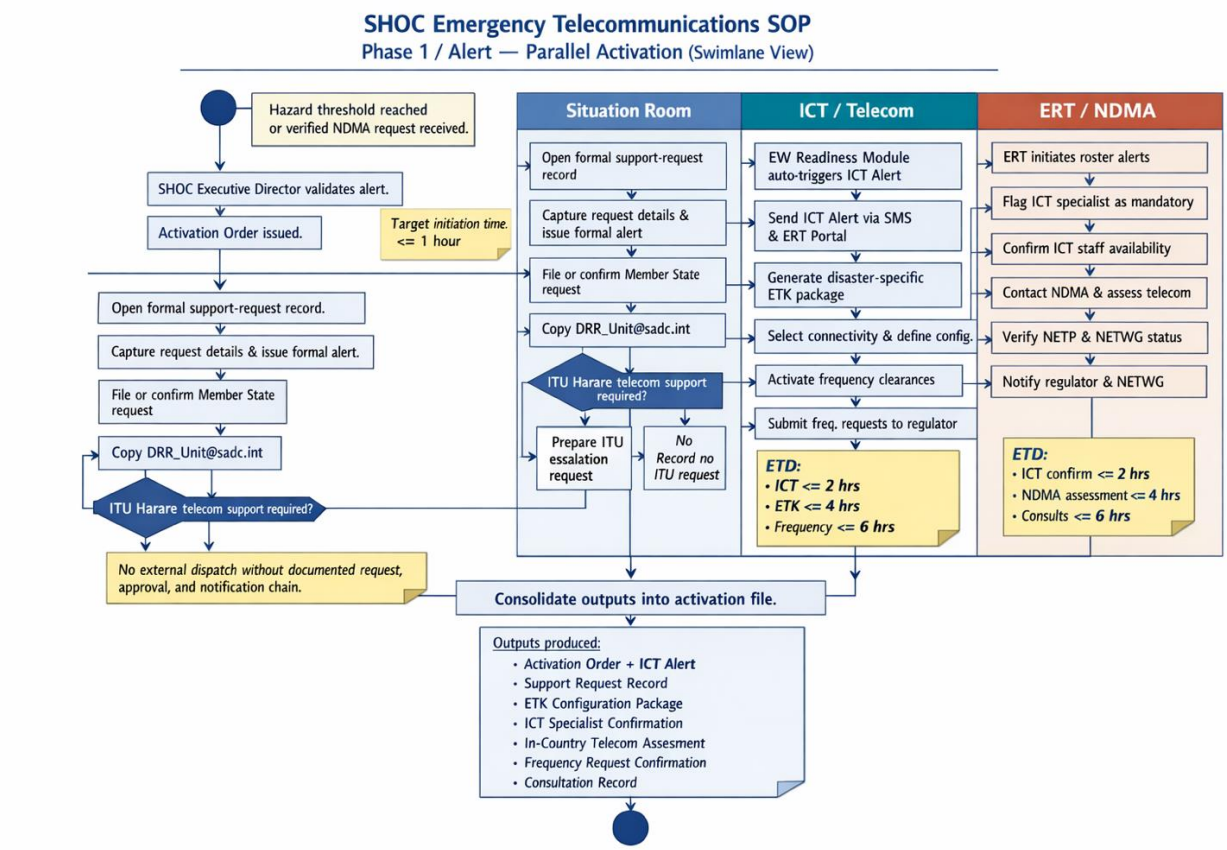


Figure 3: Parallel Activation Workflow — ICT Alert Synchronised with ERT Alert

6.1 Purpose

The Alert phase replaces the previous sequential activation model with a parallel activation architecture in which ICT and Emergency Telecommunications mobilisation commences simultaneously with, not after, the ERT roster alert. This approach aligns with the UNDAC Handbook (p. 67) requirement for rapid team assembly and with SOP Volume 4 Section 7.2,

which triggers automated alerts to eligible roster members within one hour of NDMA request validation.

6.2 Parallel Activation Workflow

When a hazard threshold is reached or a verified NDMA request is received, the following actions shall commence in parallel across three operational streams: the Situation Room issues the formal alert, the ICT and Emergency Telecommunications team activates pre-staging protocols, and the ERT Coordinator initiates roster alerts. This parallel model eliminates the sequential delay inherent in the previous workflow where telecommunications mobilisation began only after the Activation Order. Where Member State support is requested, the formal documentary request pathway shall be initiated in parallel through registry@sadc.int with copy to DRR_Unit@sadc.int, while requests for ITU Harare emergency telecommunications equipment shall be prepared against the separate ITU escalation channel and recorded in the activation file. ITU Harare Office: itu-harare@itu.int; Phone: +263 242 369015-6.

6.3 Step-by-Step: Alert Phase Procedures

6.3.1 Formal Request Routing and Support Trigger Control

For all support actions requiring regional validation, the Situation Room shall open a formal request-routing record containing the requesting Member State, requesting authority, date/time, incident type, support category, required equipment or service, documentary evidence submitted, approvals obtained, and escalation thresholds. No external dispatch to a Member State shall proceed without an auditable request trail, except where SHOC is performing immediate internal pre-staging pending formal confirmation.

Upon issuance of the Activation Order by the SHOC Executive Director, the ICT Early Warning Readiness Module shall simultaneously issue an automated ICT Alert to all ICT and Emergency Telecommunications personnel via SMS, email, and the ERT Portal. This alert shall include the disaster type, affected Member State, preliminary hazard-to-telecom profile match, ETK readiness status summary, and required ICT specialist availability confirmation. ICT specialists shall confirm availability within 2 hours, paralleling the ERT roster response timeline.

6.3.2 Disaster-Specific ETK Configuration

Upon activation, the Hazard-to-Telecom Profile Mapper shall automatically generate a disaster-specific ETK configuration package based on the identified hazard type and affected geographic area. This package shall specify primary and backup connectivity technologies (VSAT C-Band for cyclone/high-wind areas requiring robust earth stations; Starlink LEO for rapid deployment in remote areas; BGAN for mobile operations), antenna configurations optimised for the terrain and weather conditions, VHF/HF frequency plans matched to the host country regulatory environment, power system specifications based on climate and infrastructure availability, and OSSOC Telecom Go-Kit manifest with hazard-specific additions.

6.3.3 ICT Specialist Integration into ERT Shortlisting

The ERT Coordinator shall include at least one ICT/Telecommunications specialist in every ERT team composition, in accordance with the UNDAC Handbook (p. 71) identification of telecommunications as a core Mission Support function. The ICT specialist slot shall be flagged in the ERT Portal as a mandatory competency alongside the existing requirements for geographic expertise, language capability, and gender balance. The ICT Manager shall maintain a minimum of three deployment-ready ICT specialists on the ERT roster at all times, with current training certifications in VSAT/Starlink deployment, VHF/HF operations, network security, and OSSOC establishment procedures.

6.3.4 NDMA In-Country Telecom Assessment

Within 4 hours of the Alert, the ICT Specialist shall initiate remote contact with the affected Member State's NDMA to conduct a preliminary in-country telecommunications assessment. This assessment shall capture the status of national telecommunications infrastructure in the affected area, available local ISP and MNO connectivity options, regulatory requirements for foreign telecommunications equipment entry, NDMA existing telecommunications capacity and gaps, and potential OSSOC site telecommunications suitability. Assessment findings shall be documented in the In-Country Telecom Assessment Report (Annex 3.1-H) and used to refine the disaster-specific ETK configuration.

The assessment shall additionally verify: NETP status; NETWG availability; designated EW4All alerting authority; regulator position on temporary spectrum and roaming concessions; availability of Cell Broadcast, LB-SMS, CAP gateways, or public broadcaster fallback; customs entry constraints; emergency numbering considerations; and whether amateur radio or community stations exist as degraded-mode options.

6.3.5 Alert Phase Workflow Table

Step	Activity	How to Perform	Output / Deliverable	Responsible	ETD	SHOC Link
1.1	Activation Order & Parallel ICT Alert	Executive Director validates alert and issues Activation Order. ICT Module auto-triggers simultaneous ICT Alert to all telecom personnel.	Activation Order + ICT Alert issued	SHOC Executive Director / ICT Module	≤ 1 hour	SOP 4.1 ERT Portal Alert
1.2	Disaster-Specific ETK Configuration	Hazard-to-Telecom Mapper auto-selects optimal ETK config based on hazard type, terrain, and weather.	ETK Configuration Package	ICT Specialist	≤ 4 hours	Warehouse Manager
1.3	ICT Specialist Availability Confirmation	ICT specialists confirm availability via ERT Portal and SMS. ICT Manager confirms deployment candidate.	ICT Specialist Confirmation	ICT Manager / ERT Coordinator	≤ 2 hours	ERT Shortlist
1.4	NDMA In-Country Telecom Assessment	Remote assessment of host country telecom infrastructure, regulatory requirements, and OSSOC site suitability.	In-Country Telecom Assessment (Annex 3.1-H)	ICT Specialist	≤ 4 hours	NDMA DRR Focal Point
1.5	Frequency Pre-Clearance Activation	Activate relevant pre-clearances from cache. Submit disaster-specific frequency requests to host regulator.	Frequency Request Confirmation	ICT Manager / Legal Advisor	≤ 6 hours	Host Regulator / ITU
1.6	Formal request routing	Open the SHOC support-request record; file or confirm Member State request through: registry@sadc.int copied to DRR_Unit@sadc.int; For Requests to ITU Harare equipment request where required.	Support Request Record	Situation Room / ICT Manager	≤ 2 hours	Secretariat Infrastructure Directorate ICT Division / DRR Unit

1.7	NETWG and regulator notification	Notify NETP focal point, regulator, and NETWG; validate roaming, public warning, customs, and operator participation requirements.	Consultation Record	ICT Specialist / NDMA Focal Point	≤ 6 hours	Regulator / MNOs / NETWG

7. Phase 2: Standby — Team and Kit Preparation

7.1 Purpose

The Standby phase focuses on finalising team composition, assembling equipment, and developing the operational plan. In alignment with the UNDAC Handbook (pp. 67–69), which explicitly includes telecommunications equipment and solutions as a key logistics consideration within the Plan of Action, this phase requires the ICT Specialist to contribute a dedicated telecommunications section to the ERT PoA and to assemble the OSSOC Telecom Go-Kit for forward deployment.

7.2 Step-by-Step: Standby Phase Procedures

7.2.1 ICT Telecom Contribution to ERT Plan of Action

The ICT Specialist, working with the ERT Coordinator, shall prepare and submit a Telecommunications Section for inclusion in the ERT Plan of Action. This section shall cover the OSSOC connectivity plan including primary, secondary, and tertiary communication paths; ETK manifest with serial numbers and configuration details; frequency allocation status and any pending regulatory approvals; power system requirements and backup arrangements; information management procedures for telecommunications within the team and with counterparts; cybersecurity protocols including VPN configuration and SOC monitoring arrangements; and the SHOC-to-OSSOC handoff timeline. The telecommunications PoA section shall use the template provided in Annex 3.1-I and shall be submitted to the ERT Coordinator within 12 hours of team confirmation.

7.2.2 OSSOC Telecom Go-Kit Assembly

The ICT Specialist and SHOC Warehouse Manager shall assemble the OSSOC Telecom Go-Kit based on the disaster-specific ETK configuration generated during the Alert phase. The Go-Kit is a self-contained, rapidly deployable telecommunications package designed specifically for forward OSSOC operations, distinct from the SHOC-to-NDMA backbone equipment. The standard OSSOC Telecom Go-Kit shall include: one primary VSAT or Starlink terminal with auto-pointing antenna, one BGAN terminal as backup, two VHF handheld radios for tactical team communications, one HF radio for long-range backup, one ruggedised laptop with pre-configured NOC-lite monitoring software (SolarWinds/Zabbix lightweight instance), pre-configured VPN profiles for the host country ISP environment, UPS and portable solar charging system, complete cable kit and tool bag, and the OSSOC Connectivity Readiness Checklist (Annex 3.1-J). All kit items shall be verified against the IFRC Emergency Items Catalogue (EIC) compliance standards and packed for air transport in the ERT deployment consignment. Equipment staging shall follow a four-tier escalation model: Tier 1 SHOC internal assets only; Tier 2 SHOC plus host-state assets; Tier 3 SHOC plus ITU Harare augmentation; Tier 4 SHOC plus partner-supported surge assets through approved regional or international mechanisms.

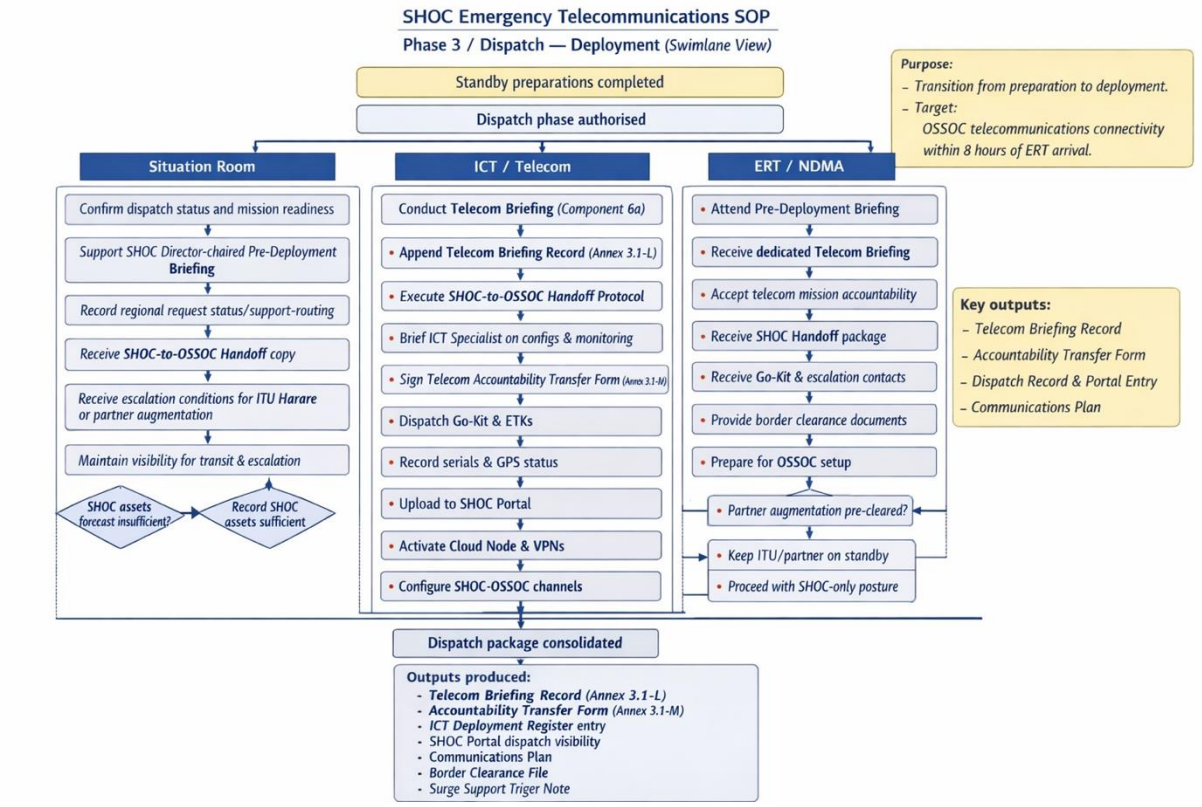
7.2.3 IFRC EIC Compliance Verification

The ICT Quality Officer shall verify all ETK and OSSOC Go-Kit equipment against the IFRC Emergency Items Catalogue specifications. This verification shall confirm power consumption compliance, weather resilience ratings for the target deployment environment, encryption capability meeting SHOC cybersecurity standards, antenna performance specifications, and deployment suitability for the identified hazard and terrain. The Verification Checklist (Annex 3.1-K) shall be completed and signed before equipment is released for dispatch. Verification shall also confirm that customs serial manifests, import/export supporting documents, and re-export documentation are prepared and matched to the actual kit to be deployed.

7.2.4 Standby Phase Workflow Table

Step	Activity	How to Perform	Output / Deliverable	Responsible	ETD	SHOC Link
2.1	Telecom PoA Contribution	ICT Specialist prepares telecommunications section for ERT Plan of Action: connectivity plan, ETK manifest, frequencies, cybersecurity.	Telecom PoA Section (Annex 3.1-I)	ICT Specialist / ERT Coordinator	≤ 12 hours post-team confirmation	ERT PoA
2.2	OSSOC Go-Kit Assembly	Assemble OSSOC Telecom Go-Kit per disaster-specific config. Pack for air transport with ERT consignment.	OSSOC Go-Kit assembled and verified	ICT Specialist / Warehouse Manager	≤ 18 hours	Logistics Unit
2.3	IFRC EIC Compliance Check	Verify all ETK and Go-Kit items against IFRC EIC specifications. Complete Verification Checklist.	EIC Verification Checklist (Annex 3.1-K)	ICT Quality Officer	≤ 24 hours	ICT Manager
2.4	Frequency Confirmation	Confirm regulatory pre-clearances activated during Alert phase. Resolve any outstanding frequency allocation issues.	Frequency Clearance Confirmation	ICT Manager / Legal Advisor	≤ 24 hours	Host Regulator
2.5	VPN & Cloud Pre-Configuration	Pre-configure VPN profiles for host country ISPs. Activate cloud mirror. Test encrypted channels.	VPN & Cloud Readiness Report	Cybersecurity Officer / Cloud Admin	≤ 24 hours	SHOC NOC
2.6	Surge equipment escalation decision	Confirm whether SHOC-only assets are sufficient or whether host-state, ITU Harare, or partner augmentation is required.	Escalation Decision Note	ICT Manager / ERT Coordinator	≤ 18 hours	Executive Director / ITU Liaison
2.7	Customs document pack	Prepare serialised manifest, temporary admission pack, border contacts, re-export forms, and facilitation letters for the dispatch file.	Customs Facilitation Pack	ICT Manager / Legal / SCM	≤ 24 hours	Border / Customs Authorities

8. Phase 3: Dispatch — Deployment



8.1 Purpose

The Dispatch phase marks the transition from preparation to deployment. In alignment with SOP Volume 4 Section 7.4, this phase includes the mandatory Pre-Deployment Briefing chaired by the SHOC Director. This Enhanced Edition adds Component 6a: a dedicated Telecommunications Briefing to the existing 9-component structure, and introduces a formal SHOC-to-OSSOC Telecom Handoff Protocol. The deployment timeline is recalibrated from 72 hours to achieve OSSOC telecommunications connectivity within 8 hours of ERT arrival.

8.2 Step-by-Step: Dispatch Phase Procedures

8.2.1 Pre-Deployment Briefing: Component 6a — Telecom Briefing

The Telecommunications Briefing shall cover the regional request status, support-routing references, OSSOC connectivity plan, frequency and licensing status, customs and border facilitation arrangements, public warning architecture, roaming and interoperability assumptions, partner augmentation trigger, cybersecurity posture, fallback communications options, and handoff accountability requirements. It shall follow the template in Annex 3.1-L and be appended to the Pre-Deployment Briefing Record.

8.2.2 SHOC-to-OSSOC Telecom Handoff Protocol

Upon completion of dispatch, telecommunications responsibility shall formally transition from the SHOC ICT team to the ERT-deployed ICT Specialist through the SHOC-to-OSSOC Telecom Handoff Protocol. This protocol requires the ICT Manager to formally brief the deploying ICT Specialist on all active configurations, pending issues, and monitoring arrangements; the ICT Specialist to sign the Telecom Accountability Transfer Form (Annex

3.1-M) accepting responsibility for all ETK and OSSOC Go-Kit items; the SHOC NOC to establish remote monitoring oversight of the OSSOC connectivity once established; and the ICT Manager to remain as primary SHOC-side contact for escalation during the transit and arrival period. The handoff shall be documented and a copy transmitted to the ERT Field Lead and the Situation Room. The handoff shall also specify which authority owns public warning approvals, which regulator and NETP focal point are on-call, and what conditions would trigger ITU Harare or other partner augmentation after arrival.

8.2.3 ETK and Go-Kit Dispatch

The OSSOC Telecom Go-Kit and supporting ETKs shall be dispatched with the ERT deployment consignment. The ICT Admin and IM Officer shall record all dispatch details in the ICT Deployment Register, including serial numbers, GPS tracking beacon activation status, and transport routing. Dispatch details shall be simultaneously uploaded to the SHOC Portal for real-time visibility. The Cloud Node shall be activated in preparation for OSSOC connectivity, and VPN profiles shall be confirmed ready for host country ISP environments. Dispatch shall not be closed until the customs facilitation pack, border escalation contacts, and re-export tracking references have been filed in the ICT Deployment Register.

8.2.4 Dispatch Phase Workflow Table

Step	Activity	How to Perform	Output / Deliverable	Responsible	ETD	SHOC Link
3.1	Pre-Deployment Briefing Telecom Component 6a	ICT Manager delivers dedicated telecom briefing: OSSOC connectivity plan, frequencies, cybersecurity, handoff, fallback procedures.	Telecom Briefing Record (Annex 3.1-L)	ICT Manager / ICT Specialist	During Pre-Deployment Briefing	ERT Field Lead
3.2	SHOC-to-OSSOC Handoff Protocol	Formal handoff briefing. ICT Specialist signs Accountability Transfer Form. SHOC NOC establishes remote oversight.	Accountability Transfer Form (Annex 3.1-M)	ICT Manager / ICT Specialist	Pre-departure	ERT Field Lead / Situation Room
3.3	ETK & Go-Kit Dispatch	Record dispatch in ICT Deployment Register. Activate GPS tracking. Upload to SHOC Portal. Activate Cloud Node.	Dispatch Record + Portal Entry	ICT Admin / IM Officer	≤ 2 hours pre-departure	Logistics Unit
3.4	Signal/Comms Groups Established	Create encrypted Signal group for ERT team. Configure SHOC-OSSOC voice/data channels. Issue call signs.	Communications Plan	ICT Specialist	≤ 1 hour pre-departure	ERT Team Members
3.5	Customs and border facilitation	Transmit customs pack, border contacts, temporary admission letters, and re-export references to convoy / airfreight handlers and receiving authorities.	Border Clearance File	ICT Admin / SCM / Legal	Pre-departure	Customs / Border Authority
3.6	External surge trigger confirmation	If SHOC assets are forecast to be insufficient, submit or pre-clear ITU Harare / partner augmentation request and document conditions for activation.	Surge Support Trigger Note	ICT Manager / Situation Room	Pre-departure	ITU Liaison / Secretariat Infrastructure Directorate (ICT Division)

9. Phase 4: On-Mission — Operations

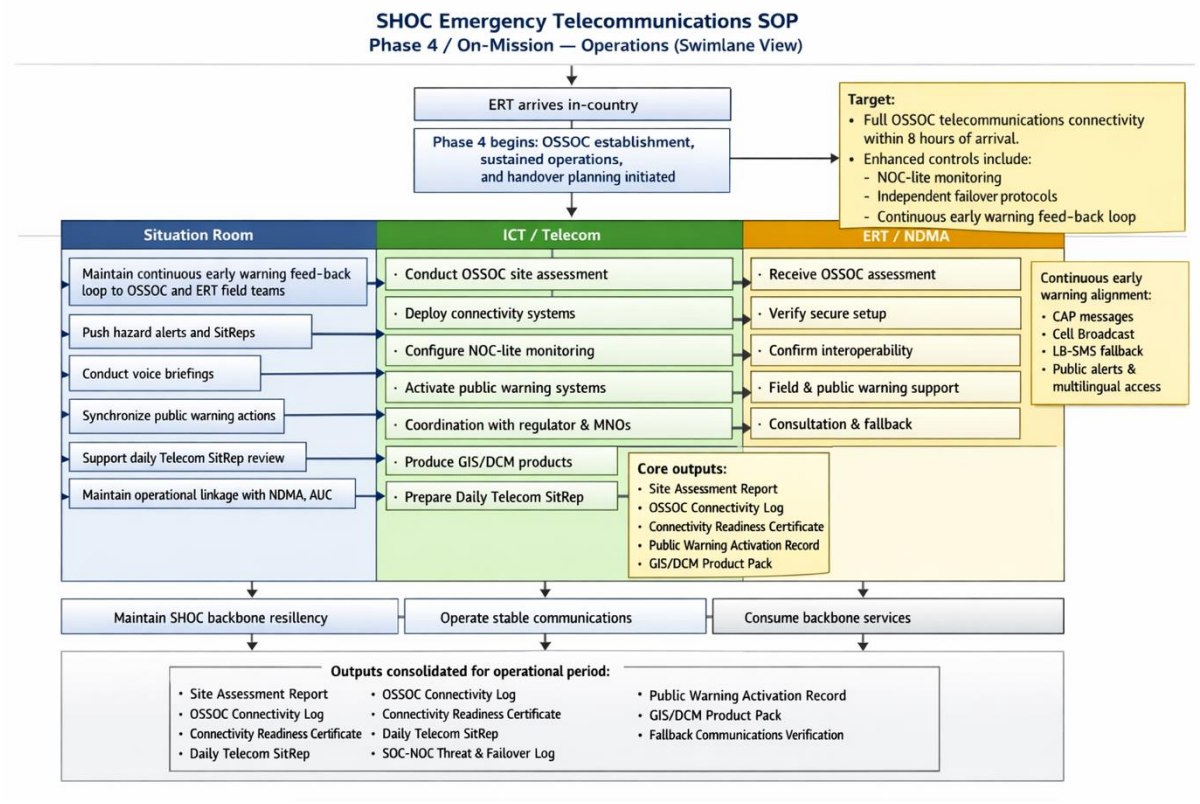


Figure 4: OSSOC On-Mission Deployment Sequence

9.1 OSSOC Telecommunications Deployment Sequence

The On-Mission phase encompasses ERT arrival, OSSOC establishment, sustained operations, and the initiation of handover planning. SOP Volume 4 Section 4.2 requires OSSOC establishment within 8 hours of arrival. This Enhanced Edition introduces dedicated OSSOC telecommunications procedures designed to achieve full connectivity within that 8-hour window, supported by NOC-lite monitoring, independent failover protocols, and a continuous early warning feed-back loop.

9.2 Step-by-Step: On-Mission Procedures

9.2.1 Arrival and Rapid Site Assessment (0–2 hours)

Upon ERT arrival in-country, the ICT Specialist shall immediately conduct a rapid telecommunications site assessment of the proposed OSSOC location. This assessment shall evaluate line-of-sight for satellite antenna pointing, local RF interference environment, available power infrastructure, physical security for equipment placement, and proximity to NDMA EOC and other coordination nodes. The assessment shall be completed within 2 hours and shall confirm or adjust the OSSOC connectivity plan developed during the Standby phase.

9.2.2 Primary Connectivity Deployment (2–6 hours)

During the 2–6 hour deployment window, the ICT Specialist shall install and align the primary VSAT or Starlink terminal; establish secure VPN connectivity to the SHOC NOC; commission backup BGAN, HF, and VHF paths; activate the tactical communications net for team movements; test data throughput, voice quality, and latency; verify interoperability with NDMA, regulator, and operator counterparts; and confirm connectivity to the SHOC Situation Room, NDMA EOC, AUC Situation Room, and any approved public warning gateway. All deployment steps shall be documented in the OSSOC Connectivity Deployment Log (Annex 3.1-N).

9.2.3 NOC-Lite Establishment and OSSOC Operational Mode (6–8 hours)

Once primary connectivity is established, the ICT Specialist shall configure the NOC-lite monitoring dashboard on the ruggedised laptop. The NOC-lite shall provide real-time monitoring of all OSSOC communication paths, automated failover triggers to backup systems (BGAN backup within 30 seconds of primary failure), bandwidth utilisation monitoring, and remote visibility for the SHOC NOC team. The ICT Specialist shall issue the OSSOC Connectivity Readiness Certificate (Annex 3.1-J) once all communication paths are verified, marking the OSSOC as fully operational. The daily Telecom SitRep cycle shall commence from this point per Annex 3.1-E.

9.2.4 Early Warning Feed-Back Loop

The SHOC Situation Room shall maintain a continuous early warning feed-back loop to the OSSOC and ERT field teams throughout the operational period. This feed-back loop shall transmit aftershock alerts, secondary hazard warnings, evolving weather patterns, upstream flood conditions, and any other hazard intelligence generated by myDEWETRA, GDACS, or regional monitoring systems. The feed-back loop shall operate via automated data push to the OSSOC NOC-lite dashboard, scheduled voice briefings (minimum twice daily) from the Situation Room to the ERT Field Lead, and emergency SMS alerts for imminent threats requiring immediate action. The ICT Specialist shall verify feed-back loop functionality daily as part of the Telecom SitRep process. The operational loop shall be synchronised with authorised public warning actions so that CAP messages, Cell Broadcast activations, LB-SMS fallback, public broadcaster alerts, and multilingual accessibility requirements can be actioned through the designated national authority without ambiguity.

9.3 SHOC Stability Communications Architecture

Public Warning, Regulatory Interoperability and GIS/DCM Operations

During on-mission operations, the ICT Specialist shall activate the agreed public warning architecture with the competent national authority, including CAP message flow, Cell Broadcast where available, LB-SMS fallback, radio and television relay, accessibility and multilingual requirements, and escalation rules for approval. In parallel, the ICT Specialist shall maintain regulator and MNO coordination for emergency SIM provisioning, national roaming activation, priority restoration, and temporary interoperability or spectrum concessions. The OSSOC shall generate GIS/DCM operational products at least once per operational period, including updated coverage maps, critical infrastructure overlays, EOC/settlement/corridor visibility, and connectivity-gap analysis for decision support.

The continuous stability and resilience of SHOC's own communications backbone is a fundamental prerequisite for all mission phases. The enhanced triple-redundancy architecture introduces three distinct redundancy layers operating simultaneously to ensure 99.9% uptime with automated failover within 30 seconds and zero data loss.

9.3.1 Integrated Cybersecurity and SOC-NOC Coordination

Cybersecurity is integrated as a core component of the SHOC stability architecture, not treated as a standalone sub-SOP. The Security Operations Centre (SOC) shall operate in continuous coordination with the Network Operations Centre (NOC), with automated threat alerts feeding directly into failover decision protocols. When the SOC detects a cyber threat (DDoS attack, intrusion attempt, malware propagation) targeting any communication path, the NOC shall automatically initiate failover to an uncompromised redundancy layer while the SOC implements containment measures. This integrated SOC-NOC approach ensures that cybersecurity incidents do not compromise SHOC backbone connectivity or OSSOC communication paths.

9.3.2 On-Mission Workflow Table

Step	Activity	How to Perform	Output / Deliverable	Responsible	ETD	SHOC Link
1	Rapid Site Assessment	ICT Specialist assesses OSSOC site: line-of-sight, RF, power, security. Confirm/adjust connectivity plan.	Site Assessment Report	ICT Specialist (ERT-deployed)	0–2 hours post-arrival	ERT Field Lead
2	Primary Connectivity Deployment	Deploy VSAT/Starlink. Establish VPN to SHOC NOC. Set up VHF/HF tactical net. Test all paths.	OSSOC Connectivity Log (Annex 3.1-N)	ICT Specialist	2–6 hours	SHOC NOC
3	NOC-Lite & Operational Mode	Configure NOC-lite monitoring. Set up auto-failover. Issue Readiness Certificate. Begin SitRep cycle.	Connectivity Readiness Certificate (Annex 3.1-J)	ICT Specialist	6–8 hours	Situation Room Dashboard
4	Early Warning Feed-Back Loop	Configure automated hazard data push to OSSOC. Verify twice-daily voice briefings. Test emergency SMS.	EW Feed-Back Loop Verification	Situation Room / ICT Specialist	8+ hours (continuous)	ERT Field Lead / NDMA
5	Daily Telecom SitRep	Prepare and submit daily SitRep per Annex 3.1-E. Upload to SHOC Portal. Update Dashboard.	Daily Telecom SitRep	ICT Specialist / IM Officer	Daily @ 18:00	SHOC Portal
6	SOC-NOC Integrated Monitoring	SOC continuously monitors for cyber threats. Automated alerts to NOC for failover decisions.	SOC-NOC Threat & Failover Log	SOC Lead / NOC Team	Continuous	SHOC ICT Manager
7	Public Warning Integration	Coordinate with the designated national alerting authority to activate CAP workflow, Cell Broadcast where available, LB-SMS fallback, broadcaster relay, multilingual messaging, accessibility checks, and approval logging.	Public Warning Activation Record	ICT Specialist / NDMA Liaison	≤ 48 hours post-OSSOC	PR Officer / Alerting Authority
8	Regulatory interoperability and roaming	Coordinate with regulator and MNOs for emergency SIMs, roaming activation, temporary spectrum/access concessions, and priority restoration for emergency telecommunications traffic.	Interoperability & Roaming Record	ICT Specialist / Regulator Liaison	≤ 24 hours post-OSSOC	MNOs / Regulator
9	GIS/DCM operational mapping	Produce coverage-gap and decision-support maps using GIS/DCM layers, including settlements, EOCs, corridors, dams, and critical infrastructure.	GIS/DCM Product Pack	ICT Specialist / IM Officer	Daily / per op period	Situation Room / NDMA
10	Community and amateur radio fallback	Confirm degraded-mode fallback using amateur radio, community stations, and local-language relay	Fallback Communications Verification	ICT Specialist / Community Liaison	≤ 48 hours post-OSSOC	NETWG / NDMA / Broadcasters

		where digital channels are compromised.				

10. Phase 5: Mission-End: Transition, Demobilisation, and Lessons Learned

10.1 Purpose

The Mission-End phase aligns with UNDAC Handbook principles that handover and exit planning should begin from the very outset of the mission, and with SOP Volume 4 SOP 4.3 which prescribes a structured five-step transition process. This Enhanced Edition introduces a telecommunications-specific Transition Protocol and mandates ICT Specialist contribution to the ERT After-Action Review, ensuring that telecom lessons are institutionalised and feed back into Phase 0 preparedness improvements.

Mission-End Workflow Table

Step	Activity	Output / Deliverable	Responsible	ETD
5.1	Transition Planning	Telecom Transition Plan (Annex 3.1-O)	ICT Specialist	14 Days Before End
5.2	Operational Handover	Operational Handover Certificate (Annex 3.1-P)	ICT Specialist & NDMA	7–3 Days Before End
5.3	Demobilisation	Telecom Demobilisation Record (Annex 3.1-Q)	ICT Specialist	Final 3 Days
5.4	After-Action Review	Telecom Lessons Learned Report (Annex 3.1-R)	ICT Specialist & MEL Officer	7–21 Days Post-Demob
5.5	Preparedness Feedback	Phase 0 Update Directive	ICT Manager	21-30 Days Post-Demob

Figure 6: Mission-End Phase

10.2 Transition Planning (14 days before mission end)

The ICT Specialist shall develop a Telecom Transition Plan (Annex 3.1-O) identifying which telecommunications assets shall be transferred to NDMA control, which shall be returned to SHOC stores, and which shall be decommissioned. The plan shall include an assessment of NDMA capacity to maintain ongoing telecommunications operations, a capacity building package for NDMA staff covering equipment operation, basic troubleshooting, and monitoring procedures, and a timeline for progressive handover of monitoring responsibilities from the OSSOC NOC-lite to the NDMA or to remote SHOC NOC oversight. Transition planning shall commence from the last third of the deployment and shall include formal closeout with the NETWG or equivalent national coordination mechanism, confirmation of whether temporary regulatory concessions will be withdrawn or normalised, and a record of remaining public warning and roaming dependencies requiring national follow-through.

10.2.2 Operational Handover (7–3 days before mission end)

The ICT Specialist shall train NDMA staff on all telecommunications systems being transferred, transfer configuration documents, frequency licences, and vendor contact details,

conduct a joint NDMA-SHOC connectivity verification to confirm that NDMA can independently operate the transferred systems, and update the SHOC asset register with all transfer records. The Operational Handover Certificate (Annex 3.1-P) shall be signed by both the ICT Specialist and the NDMA telecommunications counterpart.

10.2.3 Demobilisation (final 3 days)

The ICT Specialist shall conduct the 4-step Asset Management Protocol prescribed by SOP 4.3: reconcile all ETK and Go-Kit serial numbers against the deployment register, confirm GPS locations for all equipment, package return equipment with condition reports, and formally decommission the OSSOC telecommunications installation. All decommissioning activities shall be documented in the Telecom Demobilisation Record (Annex 3.1-Q).

10.2.4 Telecom After-Action Review Contribution (7–21 days post-demobilisation)

The ICT Specialist shall prepare a Telecom-Specific Lessons Learned Report (Annex 3.1-R) and present findings at the ERT After-Action Review facilitated under SOP 4.3 Phase 4. This report shall address ETK and Go-Kit performance against specifications, OSSOC connectivity target achievement (8-hour benchmark), triple-redundancy failover performance, early warning feed-back loop effectiveness, hazard-to-telecom profile accuracy for the deployed disaster type, regulatory and frequency allocation process efficiency, SHOC-to-OSSOC handoff protocol effectiveness, and recommendations for SOP revision. Findings shall be entered into the PMIS for trend analysis and shall directly inform Phase 0 improvements including hazard-to-telecom profile revisions, ETK pre-positioning adjustments, and quarterly drill scenario updates.

10.2.5 Mission-End Workflow Table

Step	Activity	How to Perform	Output / Deliverable	Responsible	ETD	SHOC Link
5.1	Telecom Transition Planning	Develop Telecom Transition Plan: assets for NDMA transfer vs SHOC return, NDMA capacity assessment, training plan.	Telecom Transition Plan (Annex 3.1-O)	ICT Specialist	14 days before end	ERT Field Lead / NDMA
5.2	NDMA Capacity Building	Train NDMA staff on retained systems. Transfer configs, licences, and vendor contacts.	Training Completion Record	ICT Specialist	7–3 days before end	NDMA Telecom Counterpart
5.3	Joint Connectivity Verification	NDMA-SHOC joint test of transferred systems. Confirm independent NDMA operational capability.	Operational Handover Certificate (Annex 3.1-P)	ICT Specialist / NDMA	3 days before end	SHOC NOC
5.4	Asset Reconciliation & Decommissioning	4-step Asset Management Protocol. Reconcile serial numbers, GPS locations. Package returns. Decommission OSSOC.	Telecom Demobilisation Record (Annex 3.1-Q)	ICT Specialist	Final 3 days	Logistics / Warehouse
5.5	Telecom AAR Contribution	Prepare Telecom Lessons Learned Report. Present at ERT AAR. Enter findings in PMIS.	Telecom Lessons Learned Report (Annex 3.1-R)	ICT Specialist / MEL Officer	7–21 days post-demob	ERT Coordinator / SHOC MEL
5.6	Phase 0 Feedback Loop	Feed AAR findings into hazard-to-telecom profile revisions, ETK pre-positioning adjustments, drill scenarios.	Phase 0 Update Directive	ICT Manager	21–30 days post-demob	Situation Room
5.7	NETWG / focal-point closeout	Conduct structured closeout with NETWG, NETP focal point, regulator,	National Closeout Note	ICT Specialist / NDMA	Before final exit	Secretariat Infrastructure

		and alerting authority; confirm outstanding actions for domestication, interoperability, and public warning continuity.				Director (ICT Division) / DRR Unit/ ICT Unit

11. Support of Communications Network

11.1 Purpose and Objective

SOP 3.2 ensures the maintenance, support, and resilience of SHOC's telecommunications network systems throughout all mission lifecycle phases. This Enhanced Edition integrates SOP 3.2 directly with the six-phase UNDAC lifecycle structure established in SOP 3.1, ensuring that network support activities are continuous from Phase 0 (Be Aware) through Phase 5 (Mission-End). The SOP addresses the triple-redundancy SHOC stability architecture, integrated SOC-NOC cybersecurity monitoring, automated failover systems, public alerting system integration, and OSSOC forward-deployment network support.

11.2 Enhanced Step-by-Step Process

Step 1: Deploy Network Support Teams (Enhanced)

Network support team deployment now commences at Phase 0 through continuous monitoring and readiness activities. Upon Alert activation, the ICT Officer and Assistant ICT Officer deploy teams in parallel with ERT mobilisation. Pre-positioned communication units (VSAT, Starlink, BGAN) are verified through the ETK Status Monitoring System before deployment rather than upon arrival. Teams are assigned tasks based on the disaster-specific ETK configuration generated by the Hazard-to-Telecom Profile Mapper, ensuring deployment is optimised for the specific operational scenario.

Step 2: Activate Monitoring Systems (Enhanced)

Monitoring systems (SolarWinds, Zabbix) now operate continuously under Phase 0 for SHOC backbone monitoring. Upon activation, monitoring scope is extended to include the OSSOC NOC-lite dashboard and all ERT-deployed connectivity paths. Performance metrics include

bandwidth usage, latency, error rates, failover readiness status, and SOC threat level. Alert thresholds are automatically adjusted to mission-specific parameters based on the disaster type and connectivity plan.

Step 3–4: Network Performance Assessment and Issue Resolution (Enhanced)

Network performance assessment and issue resolution are now integrated with the daily Telecom SitRep cycle. The ICT Monitoring Team conducts continuous performance assessment rather than the previous one-time evaluation at 48 hours. Issue resolution logs are shared with both the SHOC NOC and the OSSOC NOC-lite to ensure coordinated troubleshooting. The SHOC NOC maintains remote diagnostic capability for OSSOC systems, enabling SHOC-based engineers to assist with field troubleshooting via the encrypted VPN.

Step 5: Ensure Network Redundancy (Enhanced — Triple-Redundancy)

Network redundancy is enhanced from dual-carrier verification to the full triple-redundancy architecture. Layer 1 (Satellite): Primary VSAT C-Band, Backup VSAT Ku-Band, and Tertiary Starlink LEO. Layer 2 (Terrestrial): Fibre Optic primary, LTE/5G failover, and MPLS VPN inter-site. Layer 3 (Radio): HF long-range, VHF tactical, and TETRA/DMR interoperable. Automated failover target is 30 seconds. SLAs with all connectivity providers are maintained under the Frequency Pre-Clearance Cache. Failover testing is conducted quarterly under Phase 0 drills and verified during each mission activation. Supporting tools: Annex 3.2-F (Redundancy Drills & Failover Testing).

Step 6: Regulatory & Authorization (Enhanced — Pre-Clearance Model)

The pre-clearance model shall combine standing frequency and licensing readiness with incident-specific regulatory activation. It shall include regulator contact ladders, roaming concession templates, public warning gateway approvals, temporary spectrum and access concessions, import/export support letters, and disaster-specific supplementary requests, reducing the regulatory timeline from 24+ hours to confirmation within 6 hours of Alert. Supporting tools: Annex 3.2-A (Frequency Allocations & Radio Licenses).

Steps 7–10: Traffic, Inter-Agency, Public Warning, Cybersecurity (Enhanced)

Traffic prioritisation rules now include OSSOC-specific priority classes ensuring that OSSOC-to-SHOC command and control traffic receives highest priority. Inter-agency coordination procedures are enhanced with pre-negotiated MOUs maintained under Phase 0 and activated upon Alert. Public warning and community communications are integrated with the OSSOC connectivity plan, enabling the ICT Specialist to coordinate with NDMA on public alerting system integration within 48 hours of OSSOC establishment. Cybersecurity is integrated with SHOC stability through the SOC-NOC coordination model, with continuous threat monitoring feeding directly into automated failover decisions. Supporting tools: Annex 3.2-B (Traffic Prioritisation & QoS Management), Annex 3.2-C (Inter-Agency Coordination Agreements), Annex 3.2-D (Public Warning & Community Communications), and Annex 3.2-E (Cybersecurity & Data Protection).

Steps 11–17: Drills, Documentation, Scalability, Legal, Inventory, Cross-Border, Alerting (Enhanced)

Regular testing and drills are structured under Phase 0 quarterly readiness drills covering the complete activation-to-OSSOC sequence. Standard documentation templates are expanded

with new annexes for OSSOC-specific forms (Annexes 3.1-H through 3.1-R). Scalability and public telecom network integration procedures now include the OSSOC Go-Kit as a standard deployment component. The telecom infrastructure inventory is maintained through the continuous ETK Status Monitoring System. Cross-border coordination protocols are pre-established under the Frequency Pre-Clearance Cache. Inclusive public alerting systems are integrated with the OSSOC early warning feed-back loop. Supporting tools: Annex 3.2-F (Redundancy Drills & Failover Testing), Annex 3.2-G (Standard Documentation Templates), Annex 3.2-H (Scalability & Public Network Integration), and Annex 3.2-I (Post-Incident Review & Lessons Learned). Fit-for-purpose implementation also requires explicit NETWG coordination, GIS/DCM product management, controlled focal-point registers, public warning governance, customs facilitation, external surge decision logic, and sustainability tracking.

Step 18: Public Warning System Governance

Support of Communications Network shall maintain a governed public warning architecture covering CAP message exchange, Cell Broadcast enablement where available, LB-SMS fallback, emergency numbering dependencies, multilingual and accessible dissemination, testing schedules, and approval logs tied to the national alerting authority.

Step 19: GIS and Disaster Connectivity Map Operations

GIS/DCM operations shall maintain mandatory layers for radio, TV, mobile, fixed, terrestrial, satellite, maritime, and aeronautical coverage; vulnerability and settlement layers; dams; EOCs; border points; and corridor routes, with update frequency defined for preparedness and response.

Step 20: Controlled Registers and NETWG Coordination

Controlled operational registers shall be maintained for NETP focal points and alternates, EW4All alerting authorities, regulators, MNO emergency contacts, broadcasters, customs points, border officials, and ITU support contacts. NETWG notification and consultation shall be embedded in readiness, alert, response, and transition activities.

Step 21: External Surge Equipment and Customs Facilitation (New)

Network support operations shall use a documented surge model for SHOC-only, host-supported, ITU-augmented, and partner-supported deployments, with customs facilitation packs, serialised manifests, temporary admission, and re-export controls prepared before dispatch.

Step 22: Capacity Building and Sustainability

The ICT function shall maintain a role-based capability framework covering certifications, practical installation drills, TTX participation, refresher cycles, and counterpart induction. It shall also maintain annual sustainability planning for equipment refresh, subscriptions, airtime, cloud services, drills, and partner co-financing mechanisms required to keep the capability operational.

- END OF SOP -

Enhanced Risk Analysis Matrix

The following risk analysis matrix has been updated to include new risks identified through the gap analysis and to reflect the enhanced mitigation controls provided by the six-phase lifecycle model. New entries are marked with an asterisk (*).

#	Risk Category	Description	L	I	Inher.	Resid.	Mitigation / Control
1	Infrastructure Failure	Hardware or antenna malfunction during deployment	3	4	12	6	Spare routers & VSATs; daily diagnostics; SHOC NOC remote support; triple-redundancy Layer 1–3.
2	Power Outage	Loss of power to VSAT hub or switches	3	3	9	4	Hybrid solar/generator; battery banks (≥24h); UPS; OSSOC Go-Kit portable power.
3	Cyber Intrusion	Malware attack on VPN or cloud nodes	2	5	10	4	Integrated SOC-NOC monitoring; 2FA; firewall; automated threat-based failover.
4	Data Loss	Backup failure or human error	2	5	10	4	Hourly cloud backups; restricted edit rights; version control; RPO=0 replication.
5	Frequency Licence Delay	Administrative hold-up in radio licensing	2	5	10	4	Pre-Clearance Cache; ITU/SADC MOUs; pre-licensed VSAT; Alert-phase activation.
6*	OSSOC Connectivity Failure	OSSOC unable to establish comms within 8h	2	5	10	4	Dedicated Go-Kit; triple-redundancy; pre-configured VPN; ICT Specialist in ERT.
7*	EW Feed-Back Loop Failure	Situation Room unable to relay hazard updates to OSSOC	2	4	8	3	Automated data push; voice briefings 2x daily; emergency SMS; multiple paths.
8*	Handoff Accountability Gap	Unclear telecom responsibility during transit	3	3	9	3	Formal Handoff Protocol; Accountability Transfer Form; SHOC NOC oversight.
9	Personnel Safety	Hazardous conditions during deployment	3	4	12	6	Daily safety briefings; PPE; IFRC §17; medical evac plan.
10	Logistics Delay	Customs or transport bottlenecks	3	3	9	5	Customs waivers; WFP MoU; advance shipping docs; Go-Kit air-transport ready.
11	Stakeholder Coordination	Poor data sharing SHOC-NDMA	2	3	6	3	Weekly joint ops calls; CAP protocol; Situation Room briefs; EW feed-back loop.
12	Environmental Hazard	Storm damage at field site	3	4	12	6	Elevated masts; weather-proof enclosures; hazard-to-telecom profiles.
13*	ETK Readiness Degradation	Pre-positioned ETK batteries/firmware degrade	3	3	9	3	Automated ETK Status Monitoring System; dashboard alerts; 8h/24h resolution.
14*	Public Warning Approval Failure	CAP / Cell Broadcast / LB-SMS messages delayed or not authorised in time	3	5	15	6	Controlled alerting authority register; approval chain; multilingual templates; fallback broadcaster relay.
15*	Regulatory Interoperability Failure	No roaming, emergency SIM, or temporary spectrum/access concession during response	3	4	12	5	Regulator and MNO escalation ladder; pre-cleared templates; NETWG consultation; daily tracking.
16*	GIS/DCM Decision-Support Gap	Operations proceed without updated coverage and vulnerability mapping	3	4	12	5	Mandatory GIS/DCM layer pack; daily update cycle; IM ownership; validation in SitRep.
17*	Customs / Border Delay	Telecom kits delayed at entry or exit points	3	5	15	6	Customs facilitation pack; serialised manifests; border escalation contacts; re-export tracking.

18*	Sustainability Failure	Critical subscriptions, refresh, or training not funded and capability degrades	2	4	8	4	Annual readiness budget; OPEX register; partner financing review; refresh cycle tracking.

Enhanced Performance Monitoring Indicators (PMIs)

Indicator	Definition	Target	Frequency	Responsible
OSSOC Connectivity Speed	Time from ERT arrival to OSSOC full operational connectivity	≤ 8 hours	Per operation	ICT Specialist (ERT)
Deployment Speed	Time from activation to full SHOC-NDMA operational setup	≤ 48 hours	Per operation	ICT Deployment Team
Network Uptime	SHOC backbone infrastructure availability	≥ 99.9%	Continuous	NOC / ICT Office
Auto-Failover Speed	Time for automated failover between redundancy layers	≤ 30 seconds	Per failover event	NOC Team
Data Integrity (RPO)	Critical data loss events	0 events (RPO=0)	Continuous	Cloud Admin / IM Officer
System Redundancy	Active independent backup channels across 3 layers	≥ 3 layers	Quarterly	SHOC ICT
ETK Readiness Rate	Pre-positioned ETKs passing all status checks	≥ 95%	Monthly	ICT Specialist / Warehouse
EW Feed-Back Loop Latency	Time for hazard alert to reach OSSOC from Sit Room	≤ 5 minutes	Per event	Situation Room / ICT Specialist
Cross-Regional Interoperability	Successful data exchange with AUC	100%	Quarterly	SHOC Secretariat
Drill Completion Rate	Quarterly readiness drills completed vs planned	100%	Quarterly	ICT Manager / MEL
Formal Request Routing Compliance	Percentage of activations with complete support-request and approval trail	100%	Per activation	Situation Room / ICT Manager
Public Warning Activation Readiness	Percentage of operations with confirmed alerting authority, CAP chain, and fallback channels	100%	Per activation	ICT Specialist / NDMA Liaison
Roaming / Interoperability Resolution Time	Time to secure emergency SIM, roaming, or regulator concession where required	≤ 12 hours	Per activation	ICT Manager / Regulator Liaison
GIS/DCM Product Timeliness	Time to produce the required telecom decision-support map pack after activation	≤ 12 hours	Per activation	IM Officer / ICT Specialist
Customs Facilitation Success Rate	Percentage of telecom dispatches crossing border with no preventable customs delay	≥ 95%	Per deployment	SCM / Legal / ICT Admin
Training & Certification Coverage	Percentage of designated telecom responders with current required certifications and drills	≥ 90%	Quarterly	ICT Manager / HR

Acronyms and Abbreviations

Acronym	Full Form
AAR	After-Action Review
AUC	African Union Commission
BCM	Business Continuity Management
BGAN	Broadband Global Area Network
CAP	Common Alerting Protocol
CBCC	Cross-Border Coordination Cell
CERT	Computer Emergency Response Team
dLCA	Digital Logistics Capacity Assessment
DLMS	Digital Logistics Management System
DMR	Digital Mobile Radio
DRMIMS	Disaster Risk Management Information Management System
DRMSS	Disaster Risk Management Strategy and Standards
DRR	Disaster Risk Reduction
ED	Executive Director
EIC	Emergency Items Catalogue (IFRC)
EOC	Emergency Operations Centre
ERP	Enterprise Resource Planning
ERT	Emergency Response Team
ERU	Emergency Response Unit (IFRC)
ETC	Emergency Telecommunications Cluster (WFP)
ETK	Emergency Telecommunications Kit
EW	Early Warning
FEMA	Federal Emergency Management Agency (United States)
GDACS	Global Disaster Alerting Coordination System
GIS	Geographic Information System
GPS	Global Positioning System
HF	High Frequency
ICT	Information and Communications Technology
IDS	Intrusion Detection System
IFRC	International Federation of Red Cross and Red Crescent Societies
IM	Information Management
INFORM	Index for Risk Management
ISP	Internet Service Provider
ITU	International Telecommunication Union

KPI	Key Performance Indicator
LEO	Low Earth Orbit
LTE	Long-Term Evolution (4G mobile standard)
MEL	Monitoring, Evaluation and Learning
MFA	Multi-Factor Authentication
MNO	Mobile Network Operator
MOU	Memorandum of Understanding
MPLS	Multiprotocol Label Switching
NDMA	National Disaster Management Authority
NDMC	National Disaster Management Centre
NETP	National Emergency Telecommunications Plan
NIFOG	National Interoperability Field Operations Guide
NOC	Network Operations Centre
OSSOC	On-Site SHOC Operations Coordination Centre
PDC	Pacific Disaster Center
PIR	Post-Incident Review
PMIS	Performance Monitoring Indicator System
PPDR	Public Protection Disaster Response
PPE	Personal Protective Equipment
QoS	Quality of Service
RACI	Responsible, Accountable, Consulted, Informed
RF	Radio Frequency
RMO	Regional Meteorological Organisation
RPO	Recovery Point Objective
RTO	Recovery Time Objective
SADC	Southern African Development Community
SATCOM	Satellite Communications
SHOC	SADC Humanitarian and Emergency Operations Centre
SIM	Subscriber Identity Module
SLA	Service Level Agreement
SMS	Short Message Service
SOC	Security Operations Centre
SOP	Standard Operating Procedure
TETRA	Terrestrial Trunked Radio
UNDAC	United Nations Disaster Assessment and Coordination
UNDP	United Nations Development Programme
UPS	Uninterruptible Power Supply
VHF	Very High Frequency
VPN	Virtual Private Network

VSAT	Very Small Aperture Terminal
WFP	World Food Programme
DCM	Disaster Connectivity Map
EW4All	Early Warnings for All
NETP	National Emergency Telecommunications Plan
NETWG	National Emergency Telecommunications Working Group
LB-SMS	Location Based SMS

Annexes and Technical Tools

The following annexes support the enhanced SOP. Existing annexes from version 1.0 are retained from the SHOC SOP Volume 3 Annexes Compendium (Document Control: SADC-SHOC-VOL3-ANNEXES-2026) and are cross-referenced accordingly.

 Cross-Reference Note: *Annexes marked "See Annexes Compendium (Vol. 3)" are published in the companion document SHOC SOP Volume 3 Annexes Compendium (SADC-SHOC-VOL3-ANNEXES-2026).*

Reference	Title	Status / Gap Closure
Annex 3.1-A	Emergency Telecom Deployment Checklist	Retained — See Annexes Compendium (Vol. 3), ANNEX 3.1-A
Annex 3.1-B	Network Topology Diagram (SHOC ↔ NDMA ↔ Field)	Retained — See Annexes Compendium (Vol. 3), ANNEX 3.1-B
Annex 3.1-C	Connectivity Verification Template	Retained — See Annexes Compendium (Vol. 3), ANNEX 3.1-C
Annex 3.1-D	Asset Tracking & Serial Register	Retained — See Annexes Compendium (Vol. 3), ANNEX 3.1-D
Annex 3.1-E	Daily Telecom SitRep Template (ERU format)	Retained — See Annexes Compendium (Vol. 3), ANNEX 3.1-E
Annex 3.1-F	End-of-Mission Report and AAR Template	Retained (Enhanced) — See Annexes Compendium (Vol. 3), ANNEX 3.1-F
Annex 3.1-G	Quarterly Readiness Drill Report Template	Closes Gap 1–3 — To be added to Compendium
Annex 3.1-H	In-Country Telecom Assessment Report Template	Closes Gap 4–5 — To be added to Compendium
Annex 3.1-I	Telecom PoA Section Template	Closes Gap 7 — To be added to Compendium
Annex 3.1-J	OSSOC Connectivity Readiness Certificate	Closes Gap 9, 13 — To be added to Compendium
Annex 3.1-K	IFRC EIC Verification Checklist	Closes Gap 9 — To be added to Compendium

Annex 3.1-L	Pre-Deployment Telecom Briefing Template — Component 6a	Closes Gap 10 — To be added to Compendium
Annex 3.1-M	Telecom Accountability Transfer Form	Closes Gap 11 — To be added to Compendium
Annex 3.1-N	OSSOC Connectivity Deployment Log	Closes Gap 13 — To be added to Compendium
Annex 3.1-O	Telecom Transition Plan Template	Closes Gap 16 — To be added to Compendium
Annex 3.1-P	Operational Handover Certificate	Closes Gap 16 — To be added to Compendium
Annex 3.1-Q	Telecom Demobilisation Record	Closes Gap 16 — To be added to Compendium
Annex 3.1-R	Telecom Lessons Learned Report Template	Closes Gap 17 — To be added to Compendium
Annex 3.1-S	Member State Readiness Matrix & Controlled Registers	Closes fit-for-purpose gaps on NETP status, focal points and authorities — To be added to Compendium
Annex 3.1-T	Formal Request Routing and Support Trigger Register	Closes fit-for-purpose request-routing gap — To be added to Compendium
Annex 3.1-U	Customs, Border Facilitation and Re-Export Pack	Closes fit-for-purpose customs gap — To be added to Compendium
Annex 3.1-V	Public Warning Activation Governance Template	Closes fit-for-purpose CAP/CB/LB-SMS governance gap — To be added to Compendium
Annex 3.1-W	GIS/DCM Telecom Operational Map Pack Template	Closes fit-for-purpose GIS/DCM gap — To be added to Compendium
Annex 3.1-X	External Surge Equipment Escalation Decision Tree	Closes fit-for-purpose surge augmentation gap — To be added to Compendium
Annex 3.1-Y	Telecom Training, Certification and Sustainability Matrix	Closes fit-for-purpose training and sustainability gaps — To be added to Compendium
Annex 3.1-Z	Transboundary Telecom Corridor Coordination Template	Closes fit-for-purpose cross-border telecom gap — To be added to Compendium

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All references follow APA 7th edition formatting.

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19. Document Control History

The following records the version history and change rationale for this SOP, establishing an audit trail from initial development through the current version.

Version	Date	Change Description
1.0	January 2026	Initial release. SOP 3.1 (Communication Infrastructure Setup) and SOP 3.2 (Support of Communications Network) published as separate operational procedures with six annexes (3.1-A through 3.1-F) and nine annexes (3.2-A through 3.2-I). Aligned to UNDAC Handbook 8th Edition and IFRC ERU SOPs.
2.0	March 2026	Enhanced Edition. Comprehensive gap remediation based on SOP 3.1/3.2 Gap Analysis Report v1.0 (February 2026). Key enhancements: (a) Restructured into six-phase UNDAC mission lifecycle (Phases 0–5); (b) Integrated OSSOC forward-deployment connectivity procedures; (c) Added 12 new annexes (3.1-G through 3.1-R); (d) Triple-redundancy architecture; (e) SOC-NOC integrated cybersecurity; (f) Early warning feed-back loop; (g) Pre-clearance regulatory model; (h) Enhanced PMIS with 10 indicators; (i) Risk matrix expanded to 13 entries; (j) Added Sections 15.1, 16, 17, 18, 19 addressing structural gaps in annexes, references, acronyms, traceability, and document control.